

Binary Search - (4)

Q1 Find out A^{th} magical no (from the set of natural nos)

↓
A no if it is divisible by B or C.

Inputs

$$B = 2$$

$$A = \underline{8}$$

$$C = 3$$

Output

* 2, 3, 4, ~~8~~, 6, ~~12~~, 8, 9, 10, ~~16~~, 12, 13, 14, 15 16 ... ∞

1 2 3 4 5 6 7 8 8 9

↑
A

$$\text{output} = \underline{12}$$

Brute Force Soln:

$$\underline{A = 8}$$

$$\underline{B = 2}$$

$$C = 100$$

worst case

1, 2, 3, 4, 5, 6, 7, ... 16 ... ∞

⇒ Upper Bound
($A * \min(B, C)$)

(_____ 8 multiples of _____ B)
 " " " 16 // less <= 16
 {
 C

cnt = 0

for (i = 1; i <= min(B, C) * A; i++) {

if (i % B or i % C == 0) {

cnt++

if (cnt == A) {

print(i),

}

}

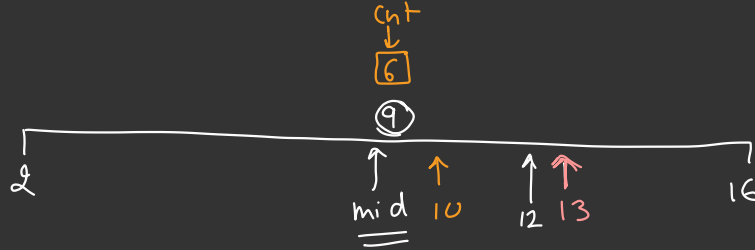
Search
space

$$S = \min(B, C)$$
$$e = A * \min(B, C)$$

$$A = 8$$

$$B = 2$$

$$C = 3$$



$ans == q \rightarrow A^h$ magical
NO
 $ans > q > A^h$ magical
NO
 $ans < q < A^h$ magical
NO

Count the no of no's $\leq q$ which are divisible by B or C

$$cnt = \frac{9}{2} + \frac{9}{3} - \frac{9}{6} = 4 + 3 - 1 = 6$$

$$\left\{ \begin{array}{l} \text{if } (cnt < A) \\ S = mid + 1 \end{array} \right.$$

$$cnt = \frac{13}{2} + \frac{13}{3} - \frac{13}{6} = 6 + 4 - 2 = 8$$

$$\left\{ \begin{array}{l} \text{elseif } (cnt == A) \{ \\ \rightarrow ans = mid; \quad // \text{ first occ where I get } cnt == 8 \\ \rightarrow e = mid - 1; \\ 3 \end{array} \right.$$

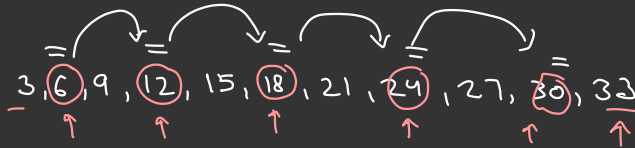
else c
 $e = mid - 1$
 }
Counting
 No

$$x = 35$$

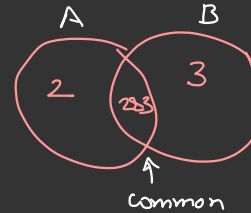
No's ≤ 35 are
 div by 2 or 3

① Multiple of 2 = $\left\lfloor \frac{35}{2} \right\rfloor = 17$

② Multiples of 3 = $\left\lfloor \frac{35}{3} \right\rfloor = 11$



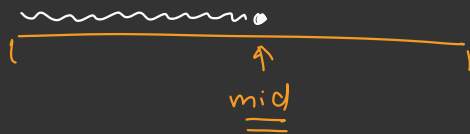
③ Multiples of $\text{lcm}(3, 2) = 6$ = $\left\lfloor \frac{35}{6} \right\rfloor = 5$



o(1) $\rightarrow$ ans = $17 + 11 - 5$
 $= 23$

2, 3, 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 21, 22, 24, 26, 27, 28, 30, 32, 33, 34

mid, B, C



$$\text{cnt} = \left\lfloor \frac{\text{mid}}{B} \right\rfloor + \left\lfloor \frac{\text{mid}}{C} \right\rfloor - \left\lfloor \frac{\text{mid}}{\text{lcm}(B, C)} \right\rfloor \quad \leftarrow$$

$$B = 4$$

$$C = 8$$

$$N = 20$$

$$\Rightarrow 4, 8, 12, 16, 20 = 5$$

$$\text{lcm}(2, 3) = 6$$

$$\left\lfloor \frac{20}{4} \right\rfloor + \left\lfloor \frac{20}{8} \right\rfloor - \left\lfloor \frac{20}{8} \right\rfloor = 5$$

$\log(\text{Range})$
 $+ \log(B, C)$

$$S = \min(B, C)$$

$$e = A * \min(B, C)$$

$$\text{ans} = \min(B, C)$$

$$\text{lcm} = \text{lcm}(B, C)$$

$$\text{while}(S \leq e) \{$$

$$\quad \text{mid} = (S + e) / 2;$$

$$O(1) \Rightarrow \text{cnt} = \text{count}(\text{mid}, B, C, \text{lcm}); \rightarrow // \text{Todo}$$

$$\text{if}(\text{cnt} == A) \{$$

$$\rightarrow \text{ans} = \text{mid}$$

$$e = \text{mid} - 1$$

$$\text{else if}(\text{cnt} < A) \{$$

$$S = \text{mid} + 1$$

$$\text{else}$$

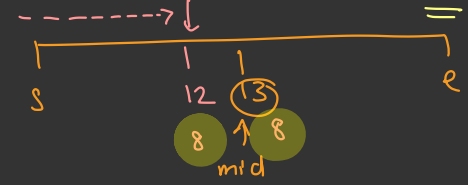
$$e = \text{mid} - 1$$

3

print(ans)

$$\left[\frac{\text{mid}}{B} + \frac{\text{mid}}{C} - \frac{\text{mid}}{\text{lcm}(B, C)} \right]$$

8^{th} Magical No
 $\uparrow$ precompute it once



$$\leq 13 \quad \frac{13}{2} + \frac{13}{3} - \frac{13}{6} = 6 + 4 - 2 = 8 = A$$

$$\leq 12 \quad \frac{12}{2} + \frac{12}{3} - \frac{12}{6} = 6 + 4 - 2 = 8 = A$$

Q2

N distinct elements in an unsorted array

Find K^{th} element if the array was sorted [1-based indexing]

arr = [2, 8, 3, 11, 14] $K=2$

$\xrightarrow{\quad}$
 2, 3, 8, 11, 14
 — — — — —
 0 1 2
 ↑
 K^{th}

① Brute force

sort(arr)
arr[k]

$O(N \log N)$

② Selection Sort ✗

$\xrightarrow{\quad K}$
 2 3 8 ...
 — — —
 N N N

$O(NK)$

Worst case $K \sim N$
 $= O(N^2)$

③

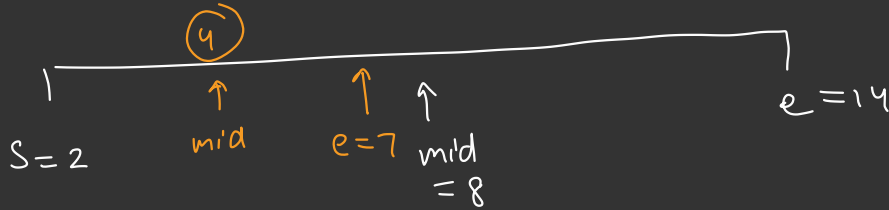
Binary Search

$$s = \min(Arr)$$

$$e = \max(Arr)$$

3, 5, 8, 2, 14

2, (3) 5, 8, 14



$k = 2$

Depends upon
Range

$$O(N \log(\max - \min))$$

while ($s \leq e$) {
 $cnt = 3$ $O(N)$
 elements less than $\leq mid$

if ($cnt > k$) {

$e = mid - 1$

}

else if ($cnt == k$) {

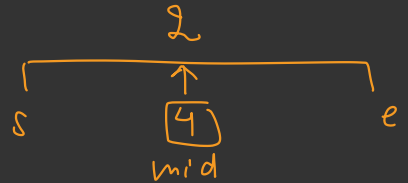
$\rightarrow ans = mid$

$\rightarrow e = mid - 1$

}

else {

cnt(4) = 2



Sorting

$N \log N$

3

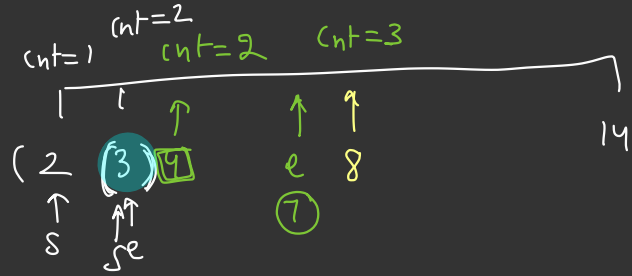
$$S = mid + 1$$

3

3, 5, 8, 2, 14

$$K = 2$$

2, 3, 5, 8, 14



$$ans = 4$$

$$mid = 2$$

Q. Given unsorted array, find k^{th} element acc (to sorted order)

10.15 

Approach based upon Quicksort idea

int Quickselect (A, s, e, k) $\Rightarrow$ 3, 6, 8, 1, 2, 9, 5

$\uparrow$ s $\uparrow$ e

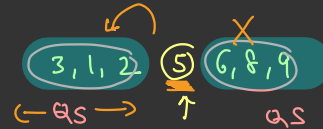
p = partition (A, s, e) Array gets Modified

if (p == k) {
 return arr[p];
}

else if (p > k) {
 return quickselect (A, s, p-1, k)
}

else {
 return quickselect (A, p+1, e, k);
}

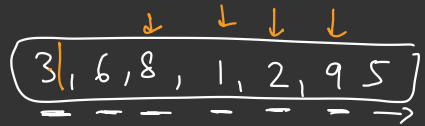
k=2



$\Rightarrow$ p=4 k==4 $\rightarrow$ 4th element

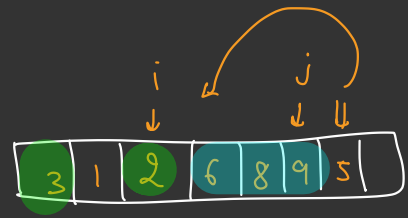
partition (arr, s, e)

L 3

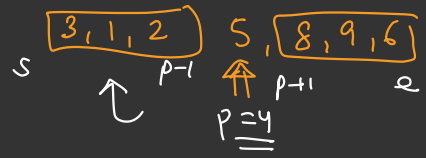


$K = 3$

1, 2, (3), 5, 6, 8, 9
 $\uparrow$
 $K = 3$

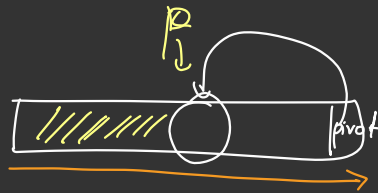


$\Downarrow$



$\uparrow$
 $p = 2 = K$

A prox
Avg case



$T(n) =$ Partition + Subproblem

$$\underline{T(n)} = \underline{cn} + \cancel{T(n/2)}$$

$$\cancel{T(n/2)} = \underline{cn/2} + \cancel{T(n/4)}$$

$$\cancel{T(n/4)} = \underline{cn/4} + \cancel{T(n/8)}$$

$\vdots$

$$\cancel{T(1)} = + \dots$$

$\log N$

$$\underline{T(n)} = cn + \underline{cn/2} + \underline{cn/4} + \dots$$

$$= cn \left(\boxed{1} + \frac{1}{2} + \frac{1}{4} + \dots \right) + \dots$$

Quicksort
2 subproblems

$$T(n) = cn + 2T(n/2) \\ \Rightarrow O(n \log n)$$

Quicksort

avg case - $O(N)$

easy
 ∞ terms

$$= kn \left(\frac{1}{1-r_2} \right)$$

$$= \cancel{2kn}$$

$$T(n) = \underline{\underline{O(n)}}$$



$$r < 1$$

$$r = \frac{1}{2} < 1$$

$$\rightarrow \frac{a}{1-r}$$

Worst
Case



$$\xrightarrow{\quad\quad\quad} \uparrow \quad cn$$

$$\begin{array}{c} 1 \ 2 \ 3 \ 4 \ (5) \\ \xrightarrow{\quad\quad\quad} \end{array} \quad + \quad c(n-1)$$

$$\begin{array}{c} 1 \ 2 \ 3 \ 4 \\ \xrightarrow{\quad\quad\quad} \end{array} \quad + \quad c(n-2)$$

+

+

$$k=1$$

$$n=6$$

$$n=5$$

$O(N^2)$ time

if array is

sorted

Divergent

5, 10, 20, - - - - - ∞

$$r = 2$$

Convergent

5, 2.5, 1.25, 0.625, - - - - ∞

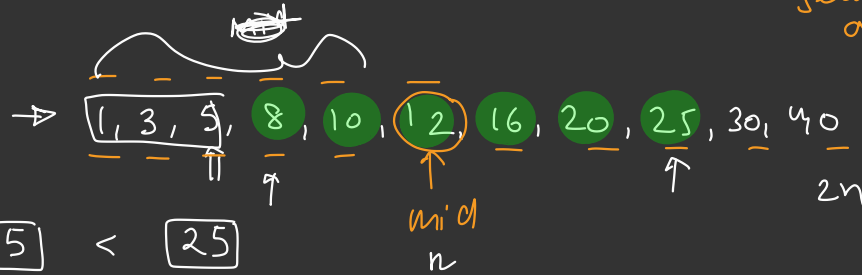
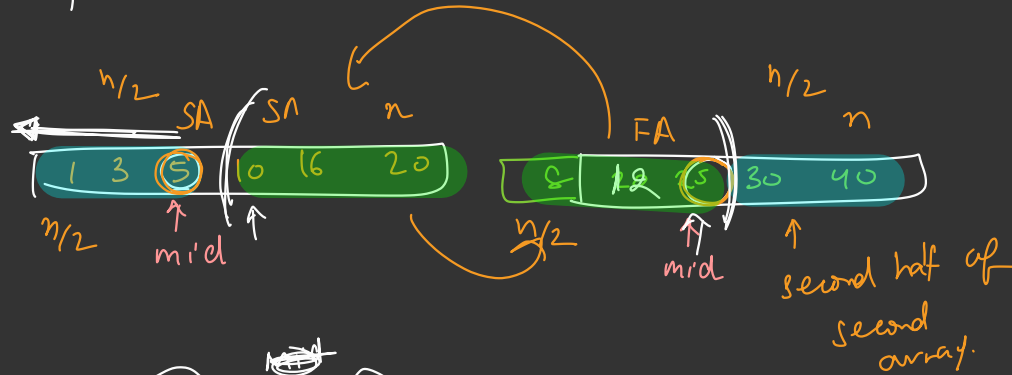
$$r = 1/2$$

$$\frac{5}{1 - 1/2} = \boxed{10}$$



The end...

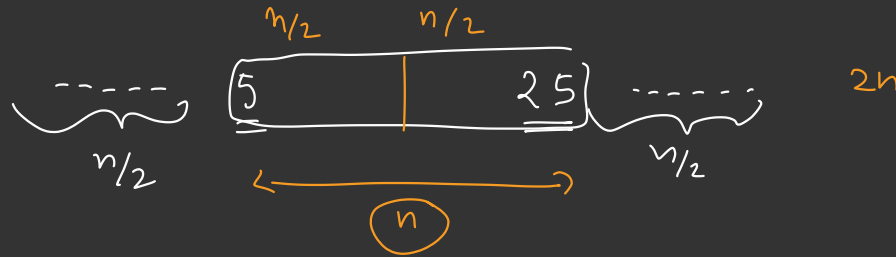
Good Night



$\boxed{11} \Rightarrow 5^{\text{th}} \text{ element}$

$$1, 3, \boxed{5} < \boxed{25}$$

first half of first array



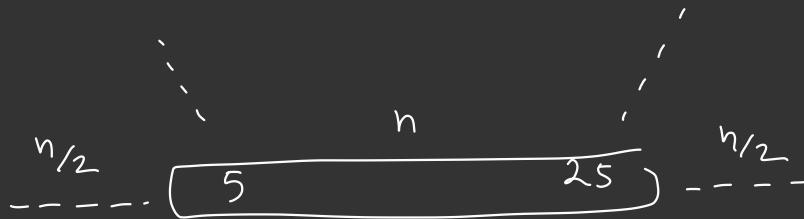
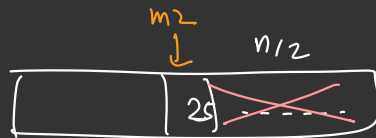
$$25 > 5$$

$2n$

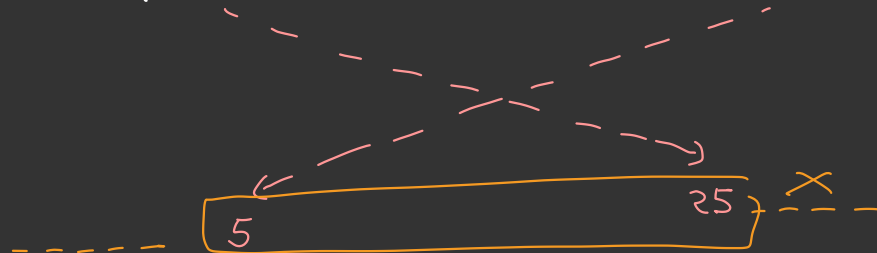
↓
 n

↓

$n/2$



$m_1 < m_2$



↑
median

